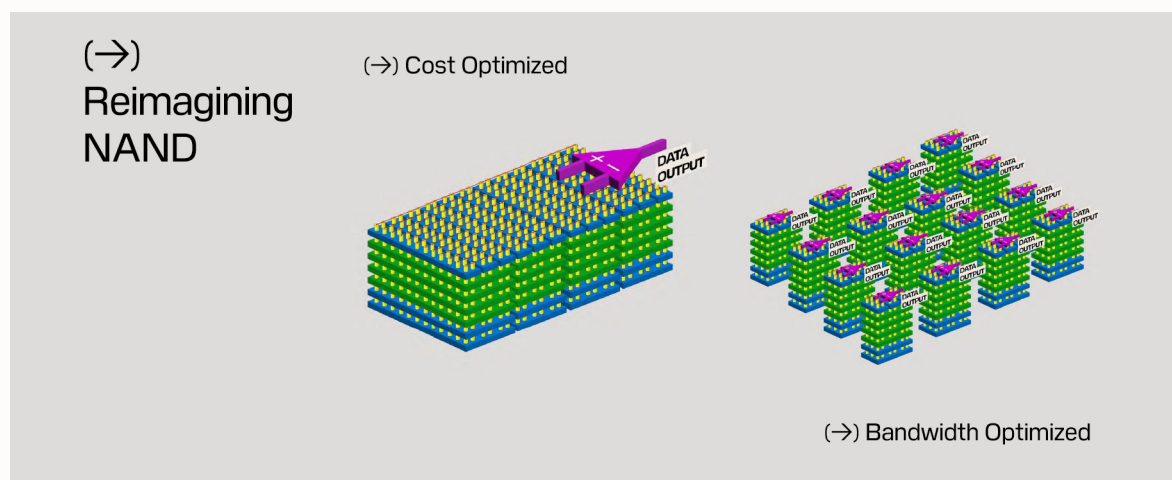


(→) SANDISK UNVEILS THE FUTURE OF MEMORY ARCHITECTURE FOR AI INTRODUCING: HIGH BANDWIDTH FLASH

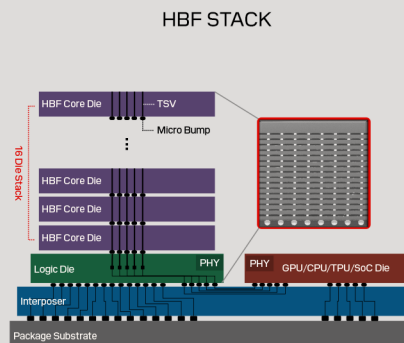
AI is pushing today's memory technologies past their limits. As AI models grow larger and more complex, inference workloads demand both massive bandwidth and significantly greater memory capacity, something traditional High-Bandwidth Memory (HBM) cannot economically or physically deliver at scale. HBM offers high speeds but is constrained by limited capacity, high power consumption, and steep costs, creating a bottleneck for developers and hyperscalers seeking to scale AI infrastructure efficiently.



To help address this critical gap, Sandisk developed a revolutionary new memory technology: High Bandwidth Flash (HBF™). HBF is built from the ground up for AI inferencing. With up to 8-16x the capacity of HBM, HBF offers industry-leading capacity at similar bandwidth and similar cost.

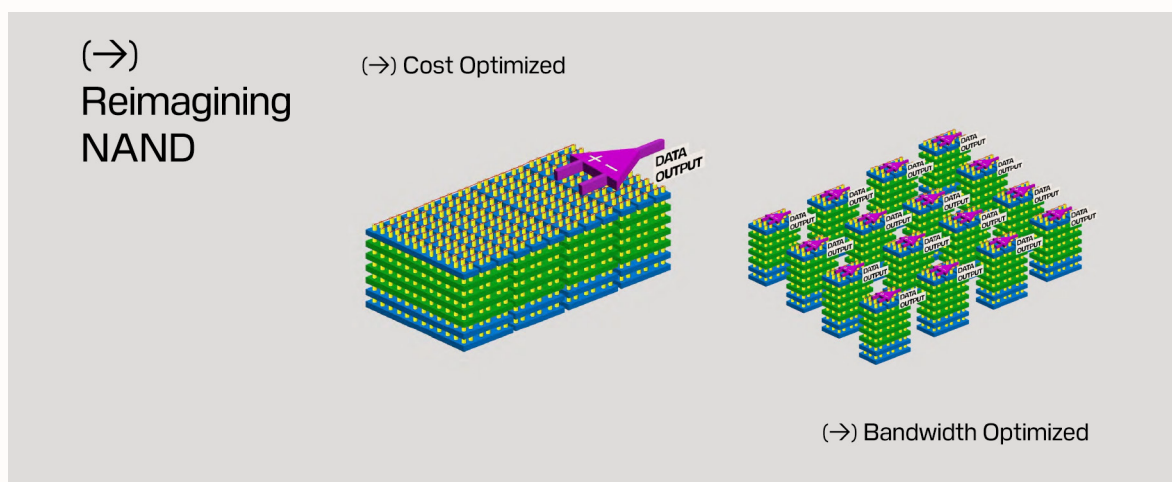
High Bandwidth Flash (HBF™) Augmenting HBM Memory with NAND Flash for AI Inference Workloads

- (→) Match HBM Bandwidth
Deliver 8-16x Capacity at Similar Cost
- (→) Enabled by BiCS Technology
With CBA Wafer Bonding
- (→) Proprietary Stacking Technology
Ultra-Low Die Warpage for 16H Stacking
- (→) Architecture Developed Over the Past Year
With inputs From Major AI Players



(→) Sandisk 公布 記憶體架構的未來：AI 引入高帶寬閃存

AI 正推動當今的記憶體技術超越其極限。隨著 AI 模型越來越大、越來越複雜，推理工作負載需要巨大的帶寬和顯著增加的記憶體容量，而傳統的高帶寬記憶體（HBM）在規模上無法經濟或物理地滿足這些需求。HBM 雖速度快，但受限於容量、功耗高和成本高，成為開發者和超大規模運營商擴展 AI 基礎設施的瓶頸。



為彌補這一關鍵缺口，Sandisk 開發了一種革命性的新型記憶體技術：高帶寬閃存（HBF™）。HBF 為 AI 推理量身打造，容量是 HBM 的 8–16 倍，提供在相似帶寬和成本下的行業領先容量。

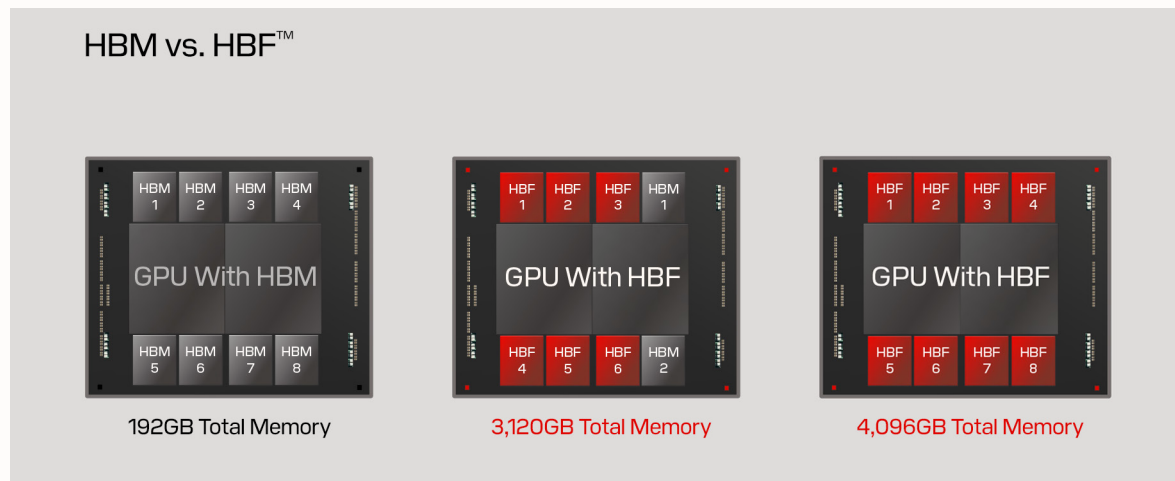
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HBF STACK

The diagram shows the HBF Stack architecture. It consists of a stack of HBF Core Dies connected by TSV (Through-Silicon Vias) and Micro Bumps. Below the HBF Core Dies is a Logic Die, which is connected to a PHY (Physical Layer) block. The PHY block is connected to a GPU/CPU/TPU/SoC Die. The entire stack is connected to an Interposer, which is mounted on a Package Substrate.

Based on NAND memory technology, HBF delivers performance within 2.2% of unlimited-capacity HBM¹, while offering a significant increase in memory capacity—unlocking AI services that traditional HBM simply can't support. Powered by Sandisk's CBA (CMOS directly Bonded to Array) technology, HBF sets a new standard for high-bandwidth memory, combining exceptional speed, density, and energy efficiency in a single architecture.

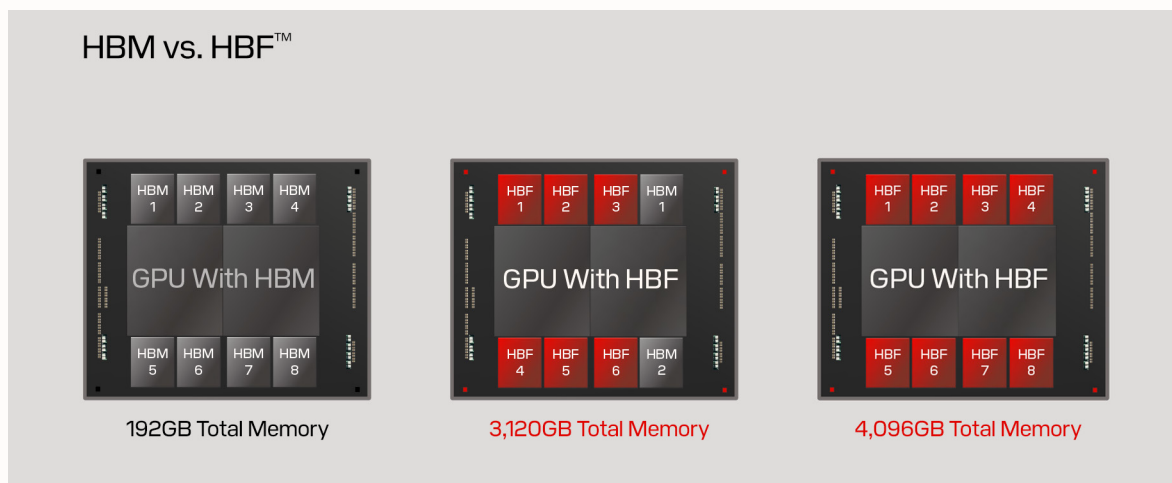


The first-generation product delivers blazing-fast read bandwidth of 1.6 TB/s and packs 256Gb per die, reaching 512GB² total capacity per 16 die stack while closely matching the physical footprint, power profile, and stack height of HBM4. HBF is purpose-built to meet the scale and intensity of next-generation AI inference workloads, offering the performance needed without compromising on capacity or efficiency.

Key highlights of HBF include:

- Our most scalable semiconductor technology in high volume manufacturing: Unlike memory solutions like DRAM, which face significant challenges on future scalability in capacity, HBF is built on Sandisk's BiCS technology, offering a clear path to greater capacity and bandwidth, as demands grow.
- High bandwidth, low cost: HBF technology offers one of the lowest cost per bit of any memory technology.
- CBA (CMOS directly Bonded to Array): Sandisk's CBA technology is a breakthrough that enables ultra-high-density, high-speed, and low-power circuits—setting a new benchmark for high-bandwidth memory performance.
- Advanced die stacking technology: HBF uses patented technology to help reduce die warpage/ stress and improve thermal conductivity, enabling 16 die stacking to deliver the targeted capacity and bandwidth.
- Non-volatile, no refresh power needed: Leveraging NAND technology, HBF retains its data even when a power source is removed and it requires no refresh power to prevent data leakage or loss.
- Thermal stability under high duress: HBF maintains thermal stability even at high temperatures expected in Data Center environments, enabling it to withstand greater operational stress without performance or reliability degradation.
- Advanced reliability and redundancy management techniques: HBF technology is designed to meet demanding requirements for concerns such as endurance and high temperature of operations in data centers and function reliably even the presence of potential failures.

基於 NAND 記憶體技術，HBF 在性能上僅比無限容量的 HBM1 差 2.2%，但在記憶容量方面有顯著提升，能支持傳統 HBM 無法達成的 AI 服務。由 Sandisk 的 CBA（CMOS 直連陣列）技術驅動，HBF 標誌著高帶寬記憶體的新標準，結合了卓越的速度、密度和能效。



第一代產品，讀取帶寬高達 1.6 TB/s，並包裝有 256Gb 晶片，16 層堆疊達到 512GB 總容量，同時在物理佔地、功耗與堆疊高度方面與 HBM4 相當。HBF 專為滿足下一代 AI 推理工作負載的規模與強度而打造，提供所需的性能，同時不犧牲容量與效率。

HBF 的主要亮點包括：

- 商用最具擴展性的半導體技術：不同於如 DRAM 的記憶體解決方案，面臨容量未來擴展的重大挑戰，HBF 建基於 Sandisk 的 BiCS 技術，能隨著需求成長，提供更大容量與帶寬的清晰路徑。
- 高帶寬、低成本：HBF 技術提供目前任何記憶體技術中成本最低的每位元成本之一。
- CBA（CMOS 直連陣列封裝）：Sandisk 的 CBA 技術是一項突破，能實現超高密度、高速與低功耗電路，為高帶寬記憶體性能樹立新標準。
- 先進的晶片堆疊技術：HBF 使用專利技術，有助於降低晶片翹曲/應力並提升熱傳導效率，實現 16 層堆疊以達到目標容量與帶寬。
- 非易失性，無需刷新電力：利用 NAND 技術，即便脫離電源，HBF 仍能保留資料，不需刷新電力以防資料洩漏或遺失。
- 高溫下的熱穩定性：HBF 在預期的資料中心高溫環境下仍維持熱穩定性，能夠承受更大的運行壓力，而不影響性能或可靠性。
- 先進的可靠性與冗餘管理技術：HBF 的設計旨在滿足資料中心在耐用性及高溫運作等要求，即使在潛在故障情況下仍能可靠運作。

HBF capabilities will only continue to improve with future generations of CBA NAND, pushing performance even further. Gen 2 and Gen 3 are expected to deliver preliminary read bandwidths exceeding 2 TB/s and 3.2 TB/s, respectively, while stack capacities are set to reach up to 1 TB and 1.5 TB³. Both are also projected to achieve greater energy efficiency, targeting 0.8x and 0.64x the power consumption of Gen 1.

High-Bandwidth Flash redefines what's possible in AI memory by building from Sandisk's low-cost, high-density CBA based NAND core technology, with architectural improvements in silicon technology, design technology and 3D stack packaging, enabling high bandwidth, high endurance, and energy efficiency. Purpose-built for the demands of AI inference, HBF delivers the speed, scale, and efficiency needed to fuel the next generation of intelligent systems—helping to solve one of AI's most critical infrastructure challenges.



¹Based on internal testing and simulation. This is simulated for reading 8-bit pretrained weights on Llama 3.1 405B parameter model. One kernel is executed on the xPU performance model at a time. Actual results may vary depending on specific circumstances and contextual factors. The comparison does not reflect the capacity advantage of HBF that can fit the full model as HBM capacity is assumed to be unlimited for modeling purposes. This demonstrates that with the higher latency and larger page size of HBF compared to HBM, the system level performance is still comparable and HBF can work for AI inference workloads.

²1GB = 1,000,000,000 bytes and 1TB = 1,000,000,000,000 bytes. Actual user capacity less.

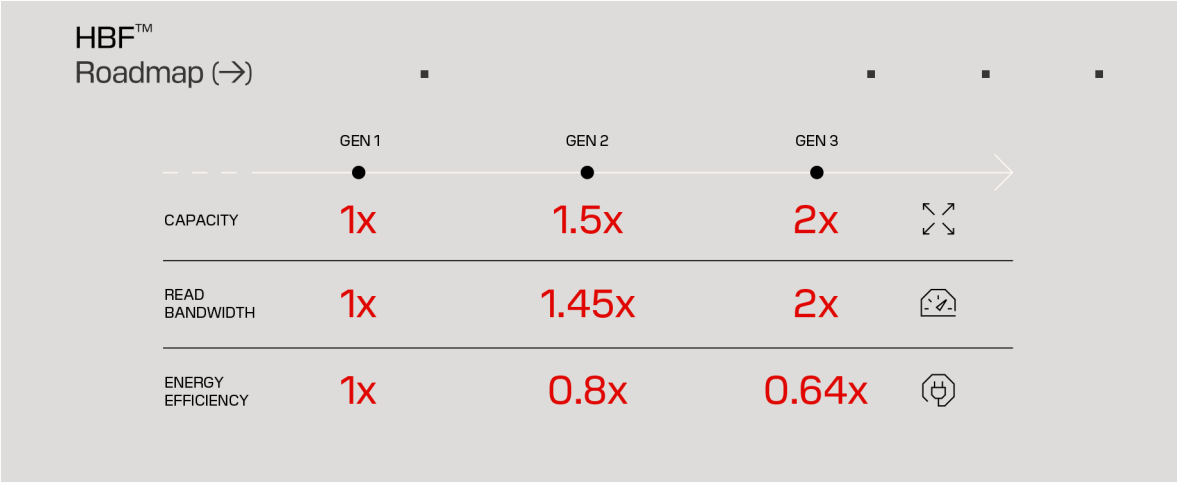
³Based on internal testing and simulation.

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隨著未來世代 CBA NAND 的推出，HBF 的能力將持續提升，性能甚至更上一層樓。預計第2代與第3代將提供超過 2 TB/s 和 3.2 TB/s 的預備讀取帶寬，堆疊容量也將達到 1 TB 和 1.5 TB。兩者都預期在能效方面取得更大進步，目標是能耗分別為第1代的 0.8 倍和 0.64 倍。高通量閃存由於採用 Sandisk 的低成本高密度 CBA 技術，並進行在硅技術、設計與 3D 堆疊封裝的架構升級，使其能提供高帶寬、高耐久度以及能效。專為 AI 推理需求打造，HBF 提供了推動下一代智慧系統所需的速度、規模與效率，有助於解決 AI 項目最關鍵的基礎設施挑戰之一。



基於內部測試與模擬。這是為了在 Llama 3.1 405B 參數模型上讀取8位元預訓練權重的模擬。每次在 xPU 性能模型上執行一個核心。實際結果可能因具體情況和背景因素而異。此比較未反映 HBF 可以容納整個模型的容量優勢，因為模擬假設 HBM 容量為無限。這展示了，即使 HBF 的延遲較高且頁面大小較大，系統層級的性能仍可媲美，且 HBF 可應用於 AI 推理工作負載。

21GB= 1,000,000,000 位元組，1TB= 1,000,000,000,000 位元組。實際用戶容量較少。基於內部測試與模擬。

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